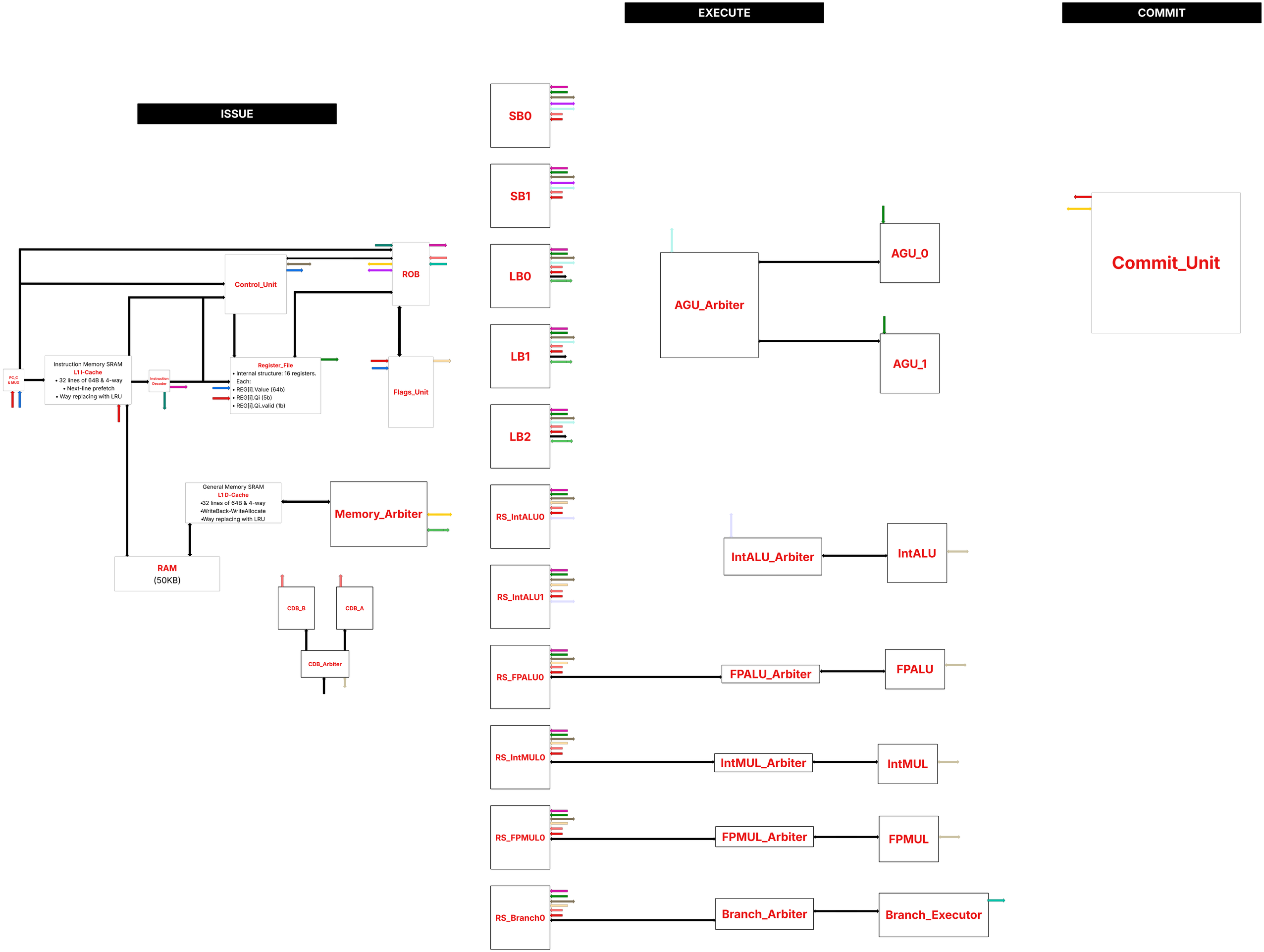
EXECUTE
COMMIT
ISSUE
SB0
SB1
LB0
LB1
LB2
RS_IntALU0
RS_IntALU1
RS_FPALU0
RS_IntMUL0
RS_FPMUL0
RS_Branch0
Control_Unit
ROB
Instruction Memory SRAM
L1 I-Cache
• 32 lines of 64B & 4-way
• Next-line prefetch
• Way replacing with LRU
PC_C & MUX
Instruction Decoder
Register_File
• Internal structure: 16 registers. Each:
• REG[i].Value (64b)
• REG[i].Qi (5b)
• REG[i].Qi_valid (1b)
Flags_Unit
General Memory SRAM
L1 D-Cache
•32 lines of 64B & 4-way
•WriteBack-WriteAllocate
•Way replacing with LRU
Memory_Arbiter
RAM
(50KB)
CDB_B
CDB_A
CDB_Arbiter
AGU_Arbiter
AGU_0
AGU_1
IntALU_Arbiter
IntALU
FPALU_Arbiter
FPALU
IntMUL_Arbiter
IntMUL
FPMUL_Arbiter
FPMUL
Branch_Arbiter
Branch_Executor
Commit_Unit